



# Engineering the facilities advanced industry runs on.

EPCM, EPCC and Energy Facility Management for cleanroom-critical industries: semiconductor, data centre, EV battery, pharmaceutical and energy. Established 1994.



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Three parts. Every section is numbered, named and opens with a one-line summary.

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# 01

## The Company

SECTIONS 01.1 TO 01.8

Established 1994 · 32 years in total facility solutions

450 people across 6 global offices

ISO 9001 · ISO 14001 · ISO 45001 certified



## About IAQ

Who we are, what we build, and the models we deliver under.

# Established 1994. A trailblazer in total facility solutions.

Engineering, procurement, construction and maintenance for the industries where contamination, uptime and safety decide the outcome, from semiconductor wafer fabs to gigafactories, data centres and pharmaceutical plants.

### Vision

To be a regional facility solutions provider with engineering excellence that facilitates technological innovation and advancement in quality of life.

### Mission

Providing innovative, sustainable facility and engineering solutions benefitting our clients and stakeholders, driven by our leadership, employees and partners globally.

### Delivery Models

**EPCM · EPCC · ENERGY FACILITY MANAGEMENT**

Engineering and construction management, full EPCC from design through commissioning, and ongoing energy and facility management.

## IAQ at a Glance

The company in six numbers, and the certifications that stand behind them.



VISUAL PLACEHOLDER

Corporate facility exterior, IAQ delivered project



32

YEARS EXPERIENCE



450

EMPLOYEES



200+

PROJECTS COMPLETED



1.5 mil m<sup>2</sup>

CLEANROOM BUILT-UP AREA



6

GLOBAL OFFICES



20 t / yr

CARBON FOOTPRINT REDUCED

### CERTIFICATION & RECOGNITION

INTERTEK ISO 9001:2015

INTERTEK ISO 14001:2015

INTERTEK ISO 45001:2018

CIDB GRADE G7

PKK GRADE G7

HIGHWIRE SAFETY GOLD AWARD

MCIEA 2024 BUILDER OF THE YEAR

## Milestones

Landmark projects from 1994 to 2026, grouped into five eras of growth. The full project list is in 02.11.



### 1994 – 2000

#### Foundation

**1994**  
Established as a cleanroom specialist

**2000**  
First decade of cleanroom delivery in Malaysia



### 2006 – 2009

#### Regional Expansion

**2006**  
(EPCC) ST Microelectronics, Class 10K cleanroom

**2008**  
(PCC) Western Digital PJ, Class 10 cleanroom

**2009**  
(GC / D&B) MEMC Ipoh, Class 1 cleanroom



### 2013 – 2017

#### Landmark Plants

**2013**  
(GC) KLCC DCC Plant, largest in Malaysia

**2016**  
(EPCC) SilTerra FAB · (EPC) Infineon MKZ Class 1K

**2017**  
(GC) Ain Medicare



### 2020 – 2022

#### Global Scale

**2020**  
(GC) Vital Healthcare

**2021**  
(GC / PCC) Robert Bosch · P Project

**2022**  
(PCC) Microsoft Data Center · (EPCM / D&B) Soitec PR1A



### 2023 – 2026

#### Advanced Technology

**2023**  
(GC / D&B) XFAB 40K Expansion

**2025**  
(PCC) 160MW hyperscale data centre · (EPCC) ESCM Germany

**2026**  
(D&B) Tata Dholera DF1 · (EPCM / HU) Micron MSH

## Global Footprint

Where we operate, and the year we arrived in each market.



**1994**

### Malaysia

Turnkey design, procurement, construction and commissioning for hi-tech clients. Selangor (HQ), Penang, Johor and Kuching.



**2020**

### Sweden

Gigafactory construction for EV battery manufacture, part of Europe's clean and digital transition.



**2023**

### Singapore

Design & build EPCM services to global semiconductor clients.



**2025**

### Germany

EPCC services following semiconductor expansion in Europe.



**2026**

### India · United States

D&B turnkey for India's first wafer fab, and US entry following localisation of advanced tech facilities.

#### ADDITIONAL OFFICES



Norway



China



France



Poland

## Core Values

Six commitments that govern how every project is run. Not words, but working rules.



01

### Safety First

Our services and works are carried out to the highest standard of safety and ethics.



02

### Quality Consistency

We take pride in the quality of solutions delivered, to achieve maximum client satisfaction.



03

### Honesty and Integrity

Our core code of conduct, fostering trust, accountability, professionalism and ethical practice.



04

### Efficiency & Proficiency

Committed to maximising available resources to achieve the best result collectively.



05

### Engineering Capabilities

Our way of working revolves around engineering principles, developing precise and sustainable solutions.



06

### Pursuit of Excellence

Devoted to excellence in all our works, providing sustainable solutions to complex challenges.

## Business Units

Three companies under one group, covering the full delivery chain.



VISUAL PLACEHOLDER

EPCC delivery on an active construction site



VISUAL PLACEHOLDER

Process critical utilities and tool installation inside a fab



VISUAL PLACEHOLDER

Energy plant and facility management operations

### 01

#### IAQ Solutions Sdn Bhd

**ENGINEERING, PROCUREMENT, CONSTRUCTION & COMMISSIONING**

EPCC from conception to operation, completion and maintenance. Every stage managed, from initial design through commissioning, so clients in hi-tech industries can bring their visions to life.

### 02

#### IAQ Utility Solutions Sdn Bhd

**PROCESS CRITICAL UTILITIES & TOTAL TOOL INSTALL**

EPCM partner for semiconductor manufacturing: process-critical utility infrastructure and total tool installation, bridging facility readiness and manufacturing start-up to accelerate fab ramp-up.

### 03

#### IAQ Energy Facility Management Sdn Bhd

**ENERGY MANAGEMENT**

Energy management solutions that optimise operations and reduce carbon footprint, keeping facilities running at the highest levels of efficiency and sustainability.

## Scope of Services

Twelve packages, self-performed and integrated under one contract.



01

### Cleanroom System

ISO 3 to ISO 8 · Class 1 to Class 100K



02

### Air Conditioning System

ACMV, make-up air, recirculation



03

### Process Utilities

CDA, PCW, PV



04

### Process Exhaust System

Scrubbed, heat and general exhaust



05

### Chemical & Gas Delivery

Specialty and bulk gases, chemical delivery



06

### Ultra Pure Water

UPW generation and distribution



07

### Waste Water Treatment

Industrial effluent and recovery



08

### Fire Protection System

Detection, suppression, compliance



09

### HT & LV Electrical System

Up to 33kV substations



10

### Civil, Structural & Architectural

Greenfield and brownfield CSA



11

### Facility Monitoring & Control

FMCS, BMS, PA



12

### Tools Hookup

Progressive tool install and hookup

## Industry Focus

Seven sectors, one discipline: contamination control under code and class.



01

### Semiconductor

Wafer fab, advanced packaging, test & assembly



02

### Data Centre

Hyperscale cooling, security, uptime



03

### EV Battery

Dry rooms, moisture control, explosion proofing



04

### District Cooling & Heating

Urban energy plant and distribution



05

### Photovoltaics

Cell and module lines, toxic material handling



06

### Pharmaceuticals & Hospitals

GMP grades, regulated environments



07

### Food & Beverage

Hygienic design, quality and safety standards



+

### Your facility next

Total facility solutions, end to end

PART 02 OF 03

# 02

## Project References

SECTIONS 02.1 TO 02.11

200+ projects completed across three regions

Over 1.5 million m<sup>2</sup> of cleanroom built

ISO 3 to ISO 8 · Class 1 to Class 100K



## Portfolio Overview

How the reference book breaks down by region, and which sections cover what.



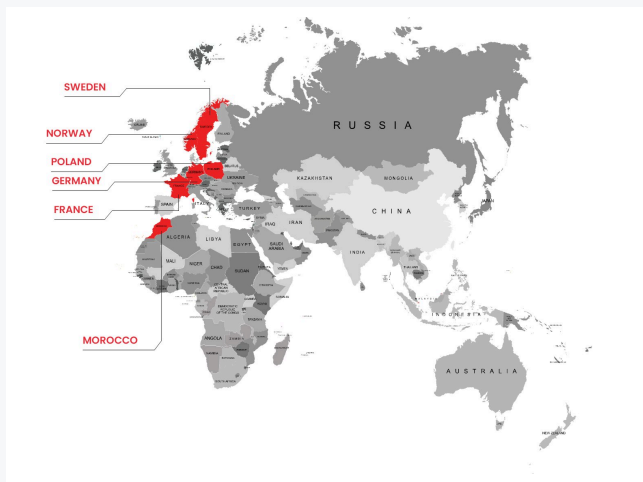
02.2 TO 02.4 · 02.8 TO 02.10

### Malaysia & Singapore

Semiconductor, data centre, pharmaceutical, photovoltaics, district cooling and energy.

**200+**

PROJECTS COMPLETED



02.5 TO 02.6

### Europe & Morocco

EV battery gigafactories, wafer fabs and advanced semiconductor in Sweden, Norway, France, Poland and Morocco.

**1.5 mil m<sup>2</sup>**

CLEANROOM BUILT



02.7

### China

Semiconductor, display, pharmaceutical, automotive and R&D facilities.

**5**

CONTRACT MODELS

GC · EPCC · EPCM · PCC · Tool Hookup

## Semiconductor · Malaysia

Wafer fab, advanced packaging, test and assembly. Further Malaysian projects are listed in 02.11.



KULIM, KEDAH · GREENFIELD WAFER FAB

### Infineon Kulim Wafer Fab 3

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| CLEANROOM   | ISO 4 · 5 · 6 · 7 (CLASS 10 · 100 · 1K · 10K)                                     |
| SCOPE       | PCC for cleanroom and mechanical works                                            |
| DESCRIPTION | General contractor for WP06, KLM3 Expansion, 35,000 m <sup>2</sup> cleanroom area |



ADVANCED PACKAGING · 75,000 M<sup>2</sup>

### P Project

|             |                                                                                          |
|-------------|------------------------------------------------------------------------------------------|
| CLEANROOM   | ISO 4 TO 7 (CLASS 10 TO 10K)                                                             |
| SCOPE       | Design & build GC: CSA, MEP, process utilities, chemical & gases system, waste treatment |
| DESCRIPTION | Progressive-built EPCM contract model for the client's advanced packaging facility       |

# Semiconductor · Singapore

Design & build EPCM in Singapore since 2023.



PHOTOMASK FACILITY

Teksend Photomask

|             |                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------|
| CLEANROOM   | CLASS 1 TO CLASS 10K                                                                                      |
| SCOPE       | EPCC for cleanroom, ACMV, electrical, process utility, BMS, PA, WWT, UPW, tool hookup                     |
| DESCRIPTION | Addition and alteration to an existing two-storey building with cleanroom facilities and ancillary office |



PROGRESSIVE TOOL HOOKUP

Micron MSH Tool HU

|             |                                                |
|-------------|------------------------------------------------|
| SCOPE       | Progressive tool hook up                       |
| DESCRIPTION | 2,000 progressive tool hookup services for MSH |

## Data Centre

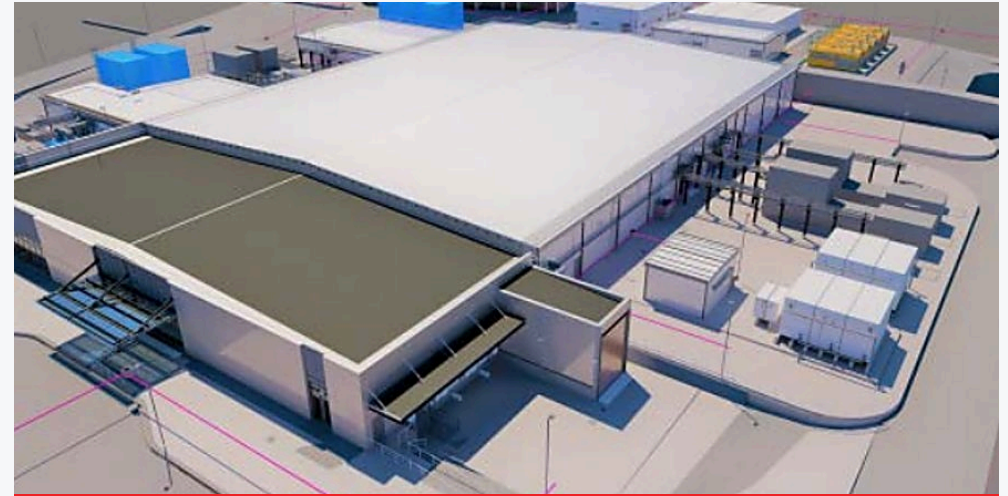
Hyperscale and colocation, 9.6 MW to 160 MW.



CLIENT CONFIDENTIAL · 160 MW

### Hyperscale Data Centre

|             |                                                                                 |
|-------------|---------------------------------------------------------------------------------|
| SCOPE       | Construction, completion, testing & commissioning for mechanical packages 1 & 2 |
| DESCRIPTION | Office, data halls and mechanical utility building                              |



KUL03 PHASE 1 · 9.6 MW

### Microsoft DTC-KUL 03

|             |                                 |
|-------------|---------------------------------|
| SCOPE       | PCC for CHW and CW piping works |
| DESCRIPTION | 9.6 MW data centre              |

# EV Battery · Europe

Dry rooms for the gigafactories: moisture control and contamination control at scale.



SKELLEFTEÅ, SWEDEN · 62,000 M<sup>2</sup>

## Northvolt AB Gigafactory

|             |                                                        |
|-------------|--------------------------------------------------------|
| ENVIRONMENT | DRY ROOM                                               |
| SCOPE       | Cleanroom and dry room architecture works              |
| DESCRIPTION | Europe gigafactory for lithium-ion battery manufacture |



NORWAY · BATTERY COAST · 8,600 M<sup>2</sup>

## Morrow Batteries

|             |                                                   |
|-------------|---------------------------------------------------|
| ENVIRONMENT | DRY ROOM                                          |
| SCOPE       | Cleanroom and dry room architectural system, ACMV |
| DESCRIPTION | Norway's Battery Coast project                    |

## Semiconductor · Europe & Morocco

Following the fabs into Europe: advanced-node cleanrooms and progressive hookup.



CLIENT CONFIDENTIAL · WAFER FAB

### 28/22nm CMOS · 16/12nm FinFET Plant

CLEANROOM CLASS 1 TO CLASS 10K

SCOPE Cleanroom & mechanical general contractor

DESCRIPTION Design, supply, installation & commissioning of cleanroom facilities



CASABLANCA, MOROCCO · 60,000 M<sup>2</sup>

### STMicroelectronics

CLEANROOM ISO 5 (CLASS 100)

SCOPE Cleanroom, mechanical, electrical, process utilities and hookup works

## Semiconductor & Display · China

EPCC across semiconductor, display and precision manufacturing.



HANGZHOU · 23,400 M<sup>2</sup>

### Ferrotec Semiconductor

CLEANROOM ISO 4 · 5 (CLASS 10 & 100)

SCOPE EPCC for cleanroom, mechanical, electrical and plumbing works and hookup



WUXI · 8,000 M<sup>2</sup>

### Infineon Technologies

CLEANROOM ISO 5 (CLASS 100)

SCOPE EPCC for cleanroom, mechanical, electrical and plumbing works and hookup

## Photovoltaics

Solar cell and module manufacturing, including toxic material and waste handling.



KULIM, MALAYSIA

### First Solar Malaysia

SCOPE PCC for mechanical & electrical, KMW building; CSA and M&E for forming gas plant; M&E for SCO1



CHINA · 57,000 M<sup>2</sup>

### Chint Solar Technology

CLEANROOM ISO 8 (CLASS 100K)

SCOPE Cleanroom system, ACMV, process utilities and tools hookup

## Pharmaceutical & Medical

GMP grades and regulated environments, from parenteral facilities to medical device plants.



GREENFIELD · MEDICAL DEVICE

**Insulet**

CLEANROOM ISO 8 (CLASS 100K)

SCOPE Design and build mechanical, cleanroom & electrical package



PUCHONG, SELANGOR

**Pharmaniaga**

CLEANROOM ISO 5 · 7 · 8 (CLASS 100 · 10K · 100K)

SCOPE Cleanroom system and ACMV for a small volume parenteral facility

## District Cooling & Energy

Plant, chillers and co-generation for urban and industrial energy.



GAS DISTRICT COOLING (M) SDN BHD

### KLCC District Cooling Plant

DESCRIPTION LARGEST DISTRICT COOLING CENTRE IN MALAYSIA

SCOPE ACMV, electrical & instrumentation, fire protection, process utility, plumbing & sanitary, water treatment, CSA



PICC PLANT, PUTRAJAYA

### GDC Putrajaya

SCOPE EPCC of plant electrification, chiller replacement and associated works at Gas District Cooling Putrajaya

## Reference Index

A selection from 200+ completed projects. The full reference list is available on request.

### MALAYSIA · SEMICONDUCTOR

XFAB Kuching 40K Expansion, 40,000 m²

SOITEC PR1A Expansion, 45,000 m²

Robert Bosch Testing Manufacturing Plant

Wafer Fab Facility & Expansion, FAB1E A/B

Texas Instruments, Melaka, 32,000 m²

Infineon Technologies, Melaka, Block 8

MEMC Ipoh, Class 1 cleanroom

STMicroelectronics, Johor

Western Digital (M), Petaling Jaya

WD Media (M), Penang, MA 28 & MA 29

Flextronics Shah Alam

### MALAYSIA · OTHER SECTORS & SINGAPORE

AUO Sunpower, Melaka, Fab 3A

Vital Healthcare, medical device plant

Ain Medicare, pharmaceutical building

M Project, food flavouring plant, 40,000 m²

P Project, MEP works, mega lab retrofit

Rapid Petronas Pengerang Co-gen

Pagoh Education Hub DCS

Rapid MCD chemical plant

Matrix (M) Sdn Bhd

T Hasegawa, spray dryer & microwave tunnel

Elegant Aura (M) Sdn Bhd

Caterpillar Asia, 7T office renovation

Micron F10A & F10NX, Singapore

National University of Singapore, tissue culture lab

### EUROPE & MOROCCO

ACC Phase 1, France, 35,000 m²

Envision, France

Soitec Semiconductor, Paris, 11,000 m²

Xfab Semiconductor, Paris

Nemotek, Rabat, Morocco

Kimoto, Poland

Sumika, Poland

### CHINA

SMIC, Ningbo, 22,000 m²

Maxtor Technology, Suzhou, 95,000 m²

All-Cent RF Technology, Wuxi, 25,600 m²

Kunshan Visionox Display, 27,000 m²

GS Magicdrive Inc., 68,000 m²

Shantou Goworld Display

M-Flex (Suzhou), 25,000 m²

MMI Industries (Wuxi)

Huawei Technologies Co., Ltd

Shanghai Roche Pharmaceuticals

Suzhou Ascentage Pharma, 25,800 m²

Jiangsu GenScript Biotech

Mindray Group Co., Ltd

Unilever · GE · Dow · Rohm and Haas R&D centres

PART 03 OF 03

# 03

## Safety, Quality & ESG

SECTIONS 03.1 TO 03.4

- Highwire Gold safety award
- ISO 45001:2018 occupational health & safety
- 20 tons per annum carbon footprint reduction



VISUAL PLACEHOLDER

Site safety in practice: crew in full PPE, toolbox briefing or EHS walkdown on an IAQ site

## ESG Commitments

Three pillars, applied to operations and to every project we take on.



### E

#### Environmental Responsibility

Minimising environmental footprint through sustainable design and construction. Energy-efficient solutions and eco-friendly technologies, prioritised at design stage.



### S

#### Social Accountability

The well-being and safety of employees and communities first. A diverse, inclusive work environment and ethical labour practices.



### G

#### Corporate Governance

Transparency, accountability and ethical decision-making, with the highest standards of compliance, building trust with clients, partners and stakeholders.

## Safety & Recognition

How our standards are independently verified and recognised.

# Judged on the things that actually matter.

MCIEA award recipients are evaluated across company performance, project management, technical expertise, innovation, quality, safety and sustainability. The same criteria our clients audit us against.

### Builder of the Year 2024

Malaysian Construction Industry Excellence Awards (MCIEA), awarded by CIDB.

### Highwire Gold

Gold-rated contractor safety performance.

### ISO 45001:2018

Occupational health and safety management, certified by Intertek.



## Closing

Your Total Facility Solutions Provider.

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# We infuse safety, quality, efficiency and excellence into every project.

From engineering design and project management to construction and maintenance, across semiconductor manufacturing, clean energy and every environment in between. IAQ is your trusted partner for total facility solutions.

## Contacts

For inquiry and business opportunity.



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